MIL-E-17764C
3 March 1982
SUPERSEDING
MIL-E-17764B
28 April 1959
(See 6.5)

MILITARY SPECIFICATION

ELECTRODE, CUTTING, TUBULAR, STEEL, COVERED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers tubular steel electrodes for use in underwater manual arc-oxygen cutting.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-W-10430 - Welding Rods and Electrodes; Preparation for Delivery of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 3112, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3439

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2.1.2 Government publications. The following Government publication forms a part of this specification to the extent specified herein.

PUBLICATION

NAVAL SEA SYSTEMS COMMAND (NAVSEA)
0929-LP-000-8010 - Welding and Cutting Underwater, Technical Manual.

(Copies of specifications, standards and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN WELDING SOCIETY (AWS)

A3.0 - AWS Definitions - Welding Terms and Definitions, Including Terms for Brazing, Soldering, Thermal Spraying, and Thermal Cutting. (DoD adopted)

(Applications for copies should be addressed to the American Welding Society, Inc., 2501 Northwest 7th Street, Miami, FL 33125.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

EW 1 - Electric Arc-Welding Apparatus. (DoD adopted)

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, NW, Washington, DC 20037.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Definitions. For the purpose of this specification, the welding terms and definitions contained in AWS A3.0 shall apply.

3.2 Material. The tubular steel core of the electrodes shall be made of a commercial grade of steel pipe or tubing.

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3.2.1 Recovered materials. Unless otherwise specified herein, all material incorporated in the products covered by this specification shall be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials.

3.3 Electrode core.

3.3.1 Dimensions.

3.3.1.1 Outside diameter (o.d.). The o.d. shall be 0.312 ± 0.002 inch.

3.3.1.2 Inside diameter (i.d.). The i.d. shall be 0.112 ± 0.008 inch.

3.3.2 Surface defects. The bore shall be uniform and free from burrs or other obstructions that may interfere with the unrestricted flow of oxygen. The ends of the electrode shall be free of sharp edges and burrs and be cut square.

3.4 Electrode length. Electrodes shall be furnished in 14-inch ± 1/8-inch tolerance end-grip straight lengths.

3.5 Coverings of electrodes.

3.5.1 Handling. Coverings shall withstand ordinary handling without damage that will affect the operation of the electrode.

3.5.2 Concentricity. The coverings on all sizes of electrodes shall be concentric to the extent that maximum core-plus-one covering dimension shall not exceed the minimum core-plus-one covering dimension by more than 5 percent of the mean dimension.

3.5.3 Uniformity. The coverings shall be such as to be consumed uniformly to produce underwater cuts as specified in 4.4.2.

3.5.4 Dielectric strength. The coverings of electrodes at room temperature and in the "dry" condition, (that is, as removed from freshly opened containers) shall have a dielectric strength sufficient to insulate effectively against a difference potential of 110 volts (V), 60 hertz (Hz), alternating current (a.c.).

3.5.5 Composition. The composition of the coverings is optional with the manufacturer, but shall be of such a nature that the electrode can be used in conjunction with oxygen to perform underwater cutting as specified in 4.4.2.

3.5.6 Stability. The stability of the coverings shall be such that electrodes, after receipt from the manufacturer, will comply with this specification after storage in original unopened containers under roof and on dry platforms for up to 6 months.

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3.5.7 Extent of coverings.

3.5.7.1 Arc ends. The arc end of each electrode shall be sufficiently bare and the covering tapered a minimum amount to permit striking of the electrode. The core shall not protrude more than 1/8-inch beyond the covering at the arc end. Chipping of the covering at the arc end may accidently occur, but when it does occur, such chipping may not exceed 1/4 the circumference of the core, and 1/4 of the circumference of the core transverse to the arc end.

3.5.7.2 Grip ends. Not less than 1-1/2 inches, nor more than 1-3/4 inches of the grip end of the electrode shall be free of covering.

3.5.8 Electrode identification.

3.5.8.1 Marking. The specification number shall be applied to the electrode covering within the space 3-1/2 inches from the grip end of the electrode.

3.5.8.2 Size and legibility. The imprinted designations shall be composed of equal-size block characters, the height of which shall be approximately 50 to 100 percent of the overall electrode diameter (core plus covering) but need not exceed a 5/32-inch height. The imprinted designations shall read from left to right from the grip-end as shown on figure 1. The color of the print shall contrast color of the electrode covering and printed designations shall be readable under normal lighting conditions.

3.5.8.3 Stability. Printed type designations shall remain discernible on electrodes and on unused portions of partially consumed electrodes or discarded stubs, and shall resist effacement by contact incidental to handling, shipping, and storing.

3.6 Usability. Electrodes shall be suitable in all respects for cutting steel plates and shapes underwater as specified in 4.4.2.

3.6.1 Operating characteristics shall remain unaffected when subjected to the salt water pressure immersion test specified in 4.4.2.1.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Electrodes selected in accordance with 4.2.1 through 4.2.4 shall be subjected to the examination and tests specified in table I.

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TABLE I. Quality conformance inspection.

Examination or test	Requirement paragraph	Inspection paragraph
Packaging and packing examination	Section 5	4.3.1
Visual and dimensional examination	3.3.1.1, 3.3.1.2, 3.3.2, 3.4, 3.5.7.1, 3.5.7.2, 3.5.8, and figure 1	4.3.2
Usability	3.5.1, 3.5.3, 3.6 and 3.6.1	4.4.2, 4.4.2.1, and table II
Concentricity	3.5.2	4.4.3
Dielectric strength	3.5.4	4.4.4 and figure 2

4.2.1 Inspection lot. For purposes of sampling for quality conformance inspection, electrodes of one size produced from one batch of covering and offered for delivery at one time shall be considered a lot.

4.2.2 Sampling for examination of packaging and packing. A random sample of shipping containers filled with packages of electrodes shall be selected by the inspector in accordance with MIL-STD-105 at inspection level S-1 and acceptable quality level (AQL) = 2.5 percent defective to verify compliance with section 5. The number of shipping containers shall be the lot size. Upon satisfactory completion of examination of filled shipping containers, a representative sample of the filled packages shall be selected by the inspector in accordance with MIL-STD-105 at inspection level S-1 and AQL = 2.5 percent defective, to verify compliance with section 5. The whole number of filled packages in the shipping containers shall be the lot size for determining the sample size. The packages selected shall be taken from shipping containers selected in the container inspection sample.

4.2.3 Sampling for examination of electrodes. A random sample of electrodes shall be selected by the inspector in accordance with MIL-STD-105 at inspection level S-1 and AQL = 2.5 percent defective to verify compliance with this specification. The whole number of electrodes in the filled packages shall be the lot size for determining the sample size. The electrodes selected in this sample shall be taken from packages selected in the packaging inspection sample.

4.2.4 Sampling for tests. A random sample of electrodes shall be selected from each inspection lot in accordance with MIL-STD-105 at inspection level S-4 for the tests specified in 4.4.2 through 4.4.4. Each electrode shall be selected from a different container.

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4.3 Examination.

4.3.1 Examination of packaging and packing. Each of the containers, both outer shipping containers and unit packages, in a sample selected in accordance with 4.2.2, shall be examined for defects of construction, for markings not as specified, and for noncompliance with the requirements specified in section 5. If the number of rejected containers exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105, the lot represented by the sample shall be rejected. Rejected lots may be resubmitted for examination provided the contractor has removed or repaired all nonconforming filled shipping containers.

4.3.2 Examination of electrodes (visual and dimensional). Each of the sample electrodes selected in accordance with 4.2.3 shall be examined by the inspector to determine conformance with this specification. Any electrode with one or more defects shall be rejected. If the number of rejected electrodes exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105, the lot represented by the sample shall be rejected. Rejected lots may be resubmitted for examination provided the contractor has removed or repaired all nonconforming electrodes.

4.4 Tests.

4.4.1 Welding equipment. Welding machines used for supply power for testing electrodes shall be variable voltage direct current (d.c.) motor-generator type or variable voltage d.c. rectifier welder, and a.c. welding transformer conforming to NEMA EW 1. The power source shall be of sufficient rating to supply current demanded by the electrode type and size under test. Open circuit voltage shall not exceed 80 V.

4.4.2 Usability. The usability of the electrodes shall be determined by making cuts underwater at a depth not greater than 10 feet on medium steel plates of 1/4, 1/2, and 1 inch in thickness. Using both a.c. and d.c., and the operating values specified in table II, the minimum rate of actual cutting and the minimum burn-off rate shall be as specified in table II.

TABLE II. Characteristics for determining usability.

Thickness of plate	Current	Voltage	Oxygen pressure	Cutting rate inches per minute	Burn-off rate linear inches of plate cut per linear inch of electrode consumed
Inch	Amperes	V	lb/in^2	Min	Min
1/4	300-325	35-40	30-40	20	1-1/2
1/2	300-350	35-40	40-50	18	1
1	300-350	35-40	50-60	15	3/4

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4.4.2.1 The electrode shall be immersed in salt water under a pressure of 100 pounds per square inch (lb/in^2) for a period of 4 hours and subsequently dried in an oven at a temperature of 90 degrees Fahrenheit (°F) for a period of 2 hours to determine compliance with 3.6 and 3.6.1.

4.4.3 Procedure for measurements of concentricity. The concentricity of the coverings on several sample electrodes shall be measured by either of the methods specified in 4.4.3.1 or 4.4.3.2 (see 4.2.4).

4.4.3.1 Concentricity. At a point near one end of the covered length of the electrode, a portion of the covering material in the form of a band approximately 1/2 inch wide shall be completely removed to the bare core, care being taken to insure that no metal is removed from the core. Using a micrometer and supplemental metal strip (approximately 1 inch long by any convenient thickness, for example, 0.125 or 0.250 inch) or a specially adapted micrometer having a "T" shaped anvil, the diameter of the core plus thickness of the covering on one side of the electrode shall be measured. The metal strip shall bridge the gap left by removal of the ring of covered material, and the total measurements shall be noted. (It is immaterial whether the thickness of the metal strip is included in the measurement.) At least five measurements shall be made on different diameters around the area from which the covering was removed, and the minimum and maximum measurements shall be noted. The maximum and minimum dimensions which show the greatest difference shall be used to determine the acceptability of the electrodes as specified in 3.5.2.

4.4.3.2 Concentricity (alternative method). The core shall be exposed by removing a small amount of covering from a spot on one side of the core near one end of the covered length, care being taken to ensure that no metal is removed from the core. The diameter of the core plus the thickness of the covering on the side opposite the bared spot shall be measured with a micrometer. The covering shall then be removed from the spot on the opposite side of the core at a point immediately adjacent to that at which the first measurement was made, and a second similar measurement made. Second and third pairs of similar measurements shall be made at approximately the midlength and the other end of the covered portion, on diameters approximately 60 degrees, respectively, from the diameter on which the first pair of measurements was made. The pair of measurements (two adjacent measurements) which shows the greatest variation shall be used to determine the acceptability of the electrodes as specified in 3.5.2.

4.4.4 Dielectric strength. Dielectric strength of coverings shall be determined by the method shown on figure 2. Methods other than the one shown, yielding the required results, may be used as alternatives if such methods are acceptable to the Government.

4.5 Certificate of conformance. A certificate of quality conformance of the examinations and tests specified in table I for each accepted lot shall be furnished to the consignee (see 6.2.2). For the usability test, actual cutting parameters (amperage, voltage and oxygen pressure) and test results (cutting rate and burn-off rate) shall be included. The minimum quality conformance data shall be as listed in figure 3 and shall be submitted on the form as shown.

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4.6 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisitions. For the extent of applicability of the requirements of referenced documents listed in section 2, see 6.4.)

5.1 Packaging. Electrodes shall be packaged level A or C, as specified (see 6.2.1).

5.1.1 Level A. Electrodes shall be packaged in class 5 containers in accordance with MIL-W-10430.

5.1.2 Level C. Electrodes shall be packaged in accordance with level C requirements of MIL-W-10430.

5.2 Packing. Electrodes shall be packed level A, B, or C, as specified (see 6.2.1), in accordance with MIL-W-10430. Palletization shall be in accordance with MIL-STD-147.

5.3 Marking. In addition to any special marking (see 5.3.1 and 6.2.1), interior packages (unit and intermediate), shipping containers, palletized and containerized unit loads shall be marked for shipment in accordance with MIL-STD-129.

5.3.1 Interior packages. Shipment marking information shall be provided on interior packages in accordance with the manufacturer's commercial practice. In addition to the lot identification, the information shall include size, specification number, manufacturer's or distributor's name, date of manufacture (month/year), and manufacturer's or distributor's brand or type designation. All packages, or the smallest integral unit within a shipping container, shall carry the following label, or equivalent, prominently displayed in legible type on the package:

"WARNING LABEL

WARNING: Protect yourself and others. Read and understand this label.

Serious injury or death can result when adequate precautions are not followed in underwater cutting operations. Personnel in charge of such operations and their operators should be thoroughly familiar with all the safety precautions contained in NAVSEA 0929-LP-000-8010.

DO NOT REMOVE THIS LABEL"

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5.3.2 Exterior shipping containers and palletized unit loads. Shipment marking information shall be provided on exterior shipping containers and palletized unit loads in accordance with the manufacturer's commercial practice. In addition to the information required for interior packages as specified in 5.3.1, the information shall include shipping destination, stock number as listed by contract, and customer's order number, customer's item number, and customer's name.

6. NOTES

6.1 Intended use. This electrode is intended for use in underwater manual arc-oxygen cutting.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Levels of packaging and packing required (see 5.1 and 5.2).
(c) Special marking (see 5.3).
(d) Quantity required (electrodes should be ordered by the pound).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraph.

Paragraph no.	Data requirement title	Applicable DID no.	Option
4.5	Documentation, Management/ Technical Support	UDI-A-26199	---

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in section 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract, regardless of whether an identical item has been supplied previously (for example, test reports).

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6.3 Inspection after delivery. Post delivery inspection of electrodes to determine conformity to this specification and for acceptance thereof is the responsibility of the consignee.

6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

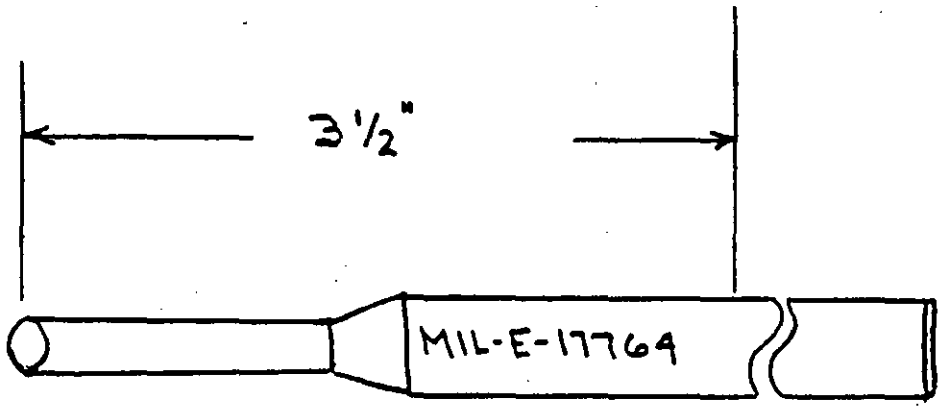
Custodians:
Navy - SH
Air Force - 99

Review activity:
DLA-GS

User activities:
Army - AT
Navy - YD

Preparing activity:
Navy - SH
(Project 3439-0442)

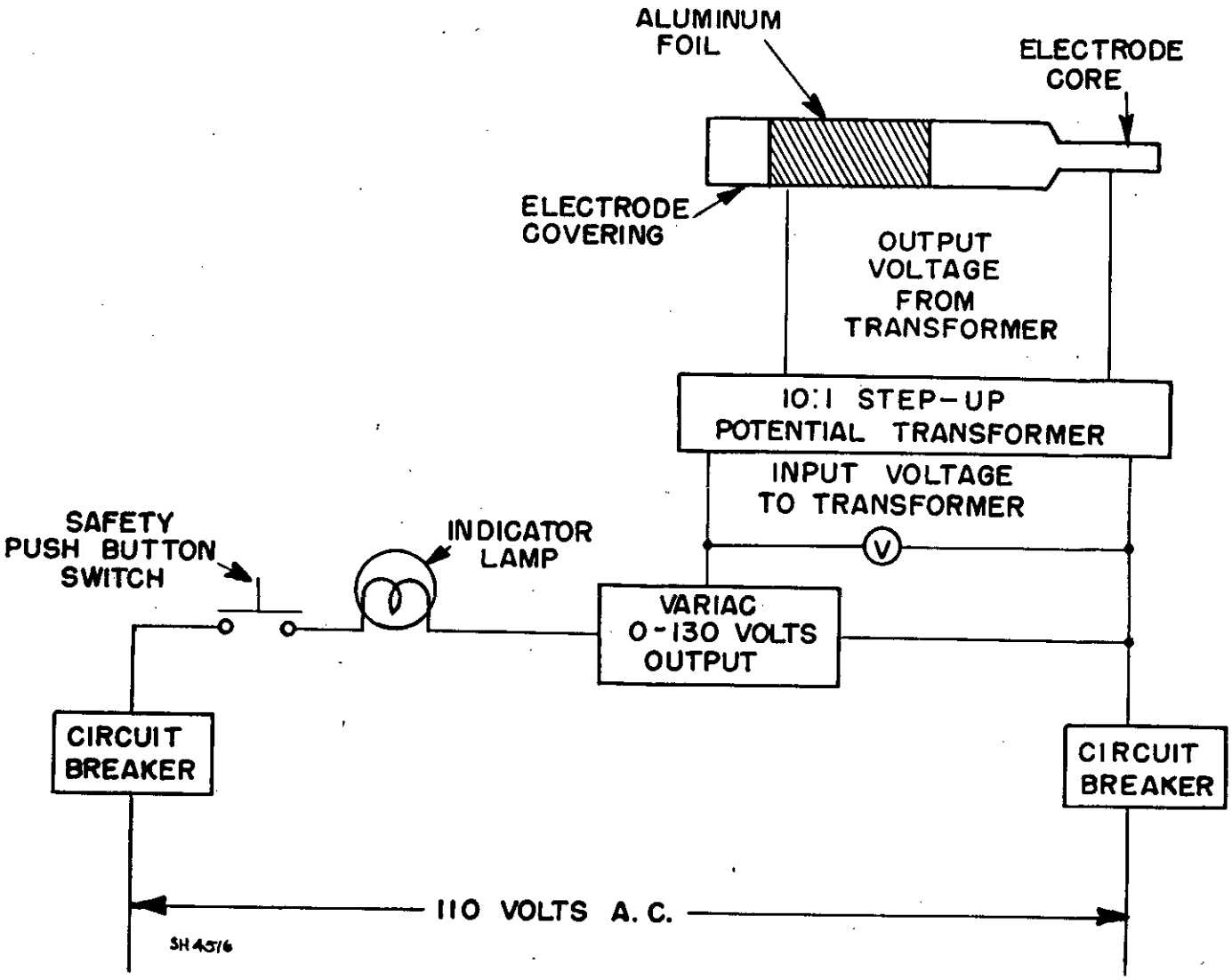
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FIGURE 1. Marking location.

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NOTES:

1. Wrap conducting aluminum foil around 6-inch length of electrode.
2. Place electrode with grip-end on one lug and foil covered section on other.
3. Close circuit breaker; press safety button and slowly rotate variac control.
4. Record maximum voltage before breakdown.
5. Dielectric strength is recorded voltage, multiplied by 10 (transformer ratio).
6. When dielectric strength is below 300 V, connect appropriate voltmeter across electrode terminals.

FIGURE 2. Circuit for dielectric strength determination.

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Manufacturer or distributor _____ Customer's name _____

Address _____ Customer's order no. _____

Date _____ Concentricity results _____

Specification MIL-E-17764 Dielectric test results _____

Diameter and length _____

Inspection level _____

Lot no. _____

Usability test - d.c. welder

Plate thickness	Amperage	Voltage	Oxygen pressure	Cutting rate	Burn-off rate

Usability test - a.c. welder

Plate thickness	Amperage	Voltage	Oxygen pressure	Cutting rate	Burn-off rate

We hereby certify that the above material has been inspected and tested in accordance with the listed specification and is in conformance with all requirements.

_____ _____

Signature of authorized company official Date

FIGURE 3. Certification of quality conformance.

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DOCUMENT IDENTIFIER AND TITLE

MIL-E-17764C

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

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